2015

DATA COMMUNICATIONS

Test 1 Time: 2 hours

Instructions to candidates

Answer all questions

Question 1

a) Explain the meaning of the following terms as used in data communication: (4 marks)

(i) Baud rate

(ii) Bit rate

(iii) Line of site transmission

(iv) Omni directional

(v) When are the baud and bit rate equal? (1 mark)

b) A wave has a frequency of 1000 Hz and a length of 250 cm, determine how fast it is moving. (2 marks)

c) What added values were/are derived from networking? (3 marks)

d) If the bit time is 8ns, what would be the data rate? (2 marks)

Question 2

a) Optic fiber is made up of three components: Name the components and explain the purpose of each. (6 marks)

b) Explain the principle of operation of optic fiber (you can use at least one diagram if you want (3 marks)

c) In the graded Index multimode, how is the grading done? (2 marks).

. What is an antenna? And how does it: (2 marks)

(i) Transmit (2 marks)

(ii) Receive signals? (2 marks)

d) Why is an isotropic antenna called idealized? (2 marks)

e) What is antenna gain? (2 marks)

Question 3

a) A wave signal has a frequency of 2 GHz and is traveling over a distance of 2.5 km. What is the anticipated signal loss due to attenuation? (6 marks)

b) A radio wave is travelling to GEO Satellite and back at a speed of $3x10^{12}$ meters per second. Calculate the time it will take o reach the receiver station (DOWN LINK) (6 marks)

c) Assuming that GEO satellites are positioned around the equator with 120 degrees separation, what is the minimum number of GEO satellites needed to cover the entire earth? (3 marks)

Question 4

a) Consider a transmission medium with a bandwidth of 2 kHz, an average signal power of 70 units and an average power of 10 units and uses 4 signal levels. Use calculations to show that the theoretical maximum transfer rates advocated for by Nyquist theorem is not achievable. (6 marks)

b)You are told that a channel is extremely noise and has the value of the signal to noise is almost zero. Calculate the channel capacity and explain your answer. (4 marks)

Question 5

A channel has a bandwidth of 500000 Hz and a signal to noise ratio of 63. What are the appropriate bit rate and signal levels? (5 marks)

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